EXPERIMENT NO: 6

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Subject Name: System Design
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Aim

To design a highly scalable video streaming platform similar to YouTube that supports heavy video uploading, adaptive bitrate streaming, and real-time social interactions with high availability and low latency.

Objectives

- Understand the architecture of a massive-scale media streaming system.
- Identify functional requirements such as direct-to-cloud uploads and personalized feeds.
- Identify non-functional requirements including CDN caching, high throughput, and latency constraints.
- Analyze the decoupling of synchronous API requests using event-driven stream processing.
- Design REST APIs for video management, search, and user interactions.

Procedure

1. Studied real-world video streaming architectures and content delivery networks.
2. Identified core entities such as Users, Videos, Interactions, and View Histories.
3. Analyzed the Pre-Signed URL pattern to bypass API Gateway bottlenecks for large file uploads.
4. Collected functional and non-functional requirements for media delivery.
5. Designed APIs for video processing, metadata retrieval, and social engagement.
6. Evaluated streaming protocols (HLS/DASH) and separated thumbnail storage for optimization.
7. Analyzed stream processing (Apache Flink/Kafka) for real-time, high-volume view aggregation.

Functional Requirements

1. System should allow users to register, log in, and manage accounts.
2. System should support uploading large raw video files efficiently.
3. System should stream video smoothly using Adaptive Bitrate Streaming based on network conditions.
4. System should support real-time social interactions (likes, dislikes, comments).
5. System should allow users to search for videos by title or metadata.

6. System should generate a personalized home page feed based on watch history.

Core Entities of the System

- Users
- Videos
- Interactions (Likes/Dislikes)
- Comments
- ViewHistory

API Design

1. Authentication API

A. **Register**: POST /api/v1/auth/register
B. **Login**: POST /api/v1/auth/login

2. Video & Feed API

A. **Initialize Upload**: POST /api/v1/videos/upload/init (Returns S3 Pre-Signed URL)
B. **Get Video Metadata**: GET /api/v1/videos/{video_id}
C. **Get Personalized Feed**: GET /api/v1/feed (Pagination Supported)
D. **Search Videos**: GET /api/v1/search?q={query}

3. Interaction API

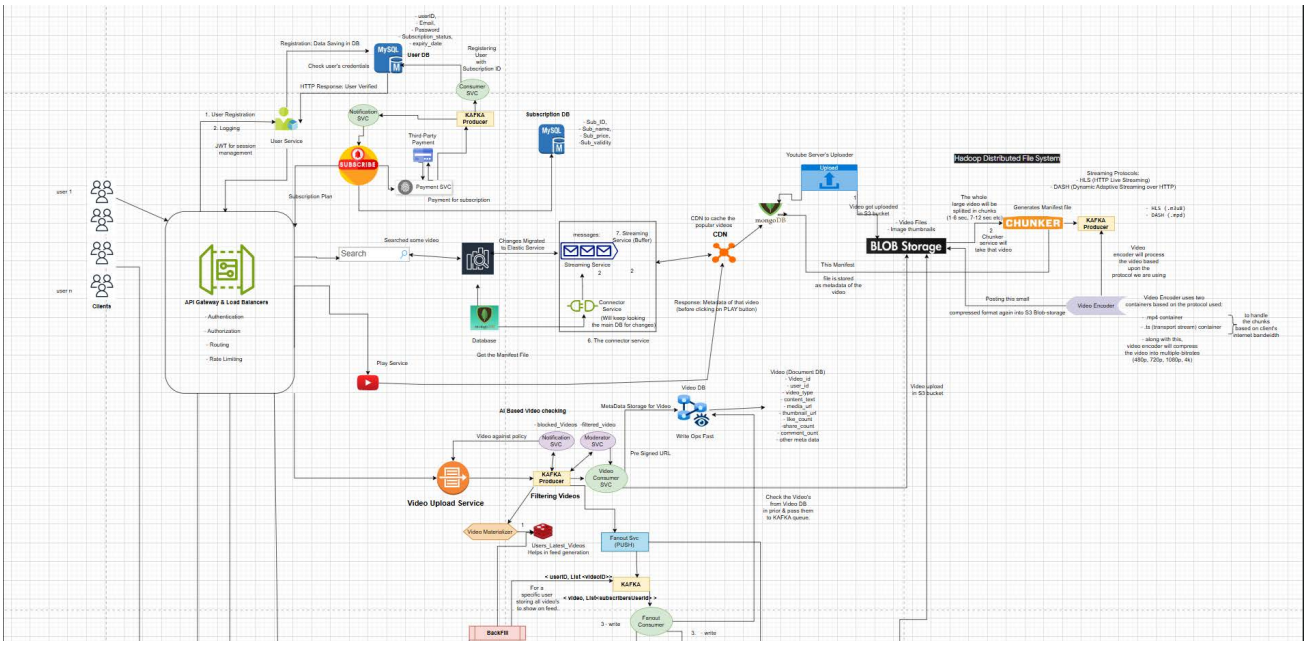
A. **Like Video**: POST /api/v1/videos/{video_id}/like
B. **Add Comment**: POST /api/v1/videos/{video_id}/comments
C. **Get Comments**: GET /api/v1/videos/{video_id}/comments

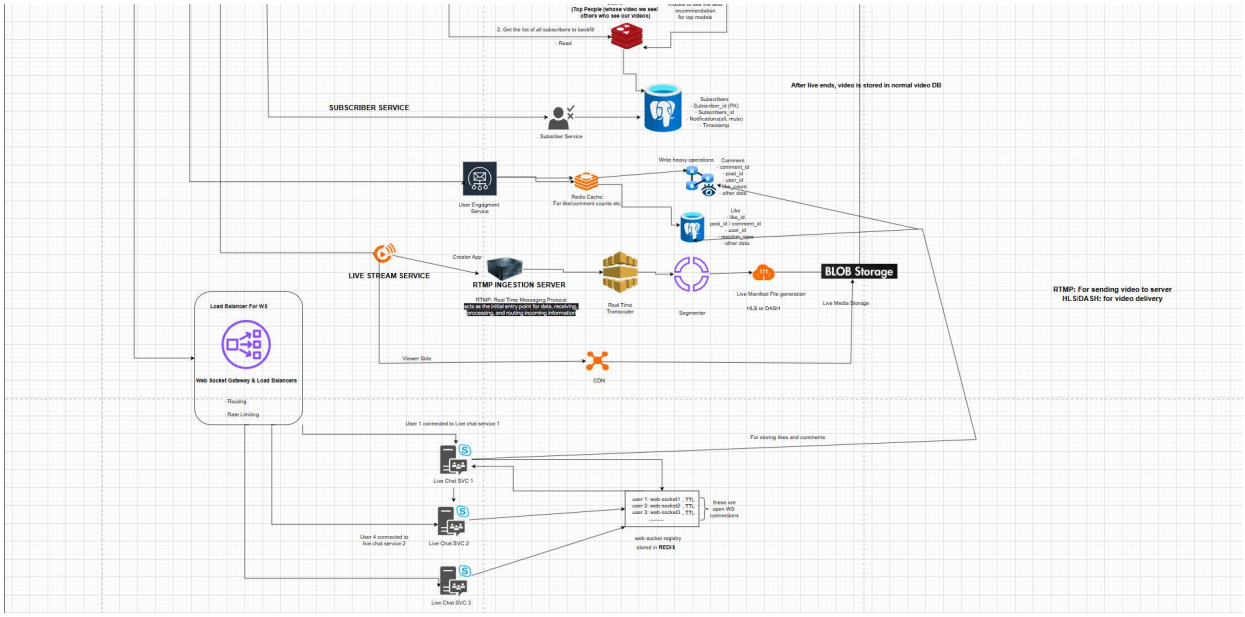
Non-Functional Requirements

1. System must handle extremely high throughput for concurrent uploads and millions of simultaneous views.
2. High Availability must be prioritized, ensuring no single point of failure.
3. System should follow an Eventual Consistency model for social metrics (view counts, likes) to optimize database write loads.
4. Video delivery latency should be minimized via Edge Caching and a Global CDN.
5. Transcoding pipeline must be horizontally scalable to handle viral upload spikes.

High Level Design (HLD)

The system consists of Client Applications, an API Gateway, and specialized microservices (Auth, Upload, Video Metadata, Feed, and Interactions). Heavy uploads bypass the backend via Pre-Signed URLs directly to Object Storage. Asynchronous transcoding pipelines process videos into HLS/DASH segments, which are delivered globally via a CDN. High-volume interactions are managed using Redis (Likes), NoSQL clusters (Comments), and a Kafka-to-Apache Flink stream processing pipeline for accurate view aggregation.


Registration: Data Saving in DB
Check user's credentials
HTTP Response: User Verified
User DB
Registering User with Subscription ID
Consumer SVC
KAFKA Producer
Notification SVC
Third-Party Payment
Payment SVC
Payment for subscription
Subscription Plan
SUBSCRIBE
User Service
1. User Registration
2. Logging
JWT for session management
Subscription DB
Sub_ID,
Sub_name,
Sub_price,
Sub_validity
user 1
user n
Clients
API Gateway & Load Balancers
Authentication
Authorization
Routing
Rate Limiting
Search
Searched some video
Changes Migrated to Elastic Service
Database
Get the Manifest File
Play Service
Streaming Service
7. Streaming Service (Buffer)
Connector Service
(Will keep looking the main DB for changes)
6. The connector service
CDN to cache the popular videos
CDN
Response: Metadata of that video (before clicking on PLAY button)
mongoDB
Youtube Server's Uploader
Upload
Video got uploaded to S3 bucket
Video Files
Image thumbnails
BLOB Storage
This Manifest
file is stored as metadata of the video
Hadoop Distributed File System
Streaming Protocols:
HLS (HTTP Live Streaming)
DASH (Dynamic Adaptive Streaming over HTTP)
The whole large video will be splitted in chunks (1-6 sec, 7-12 sec etc)
Generates Manifest file
CHUNKER
KAFKA Producer
Chunker service will take that video
Video encoder will process the video based upon the protocol we are using
Posting this small compressed format again into S3 Blob storage
Video Encoder
Video Encoder uses two containers based on the protocol used:
mp4 container
ts (transport stream) container
along with this, video encoder will compress the video into multiple bitrates (480p, 720p, 1080p, 4k)
to handle the chunks based on client's internet bandwidth
Video (Document DB)
Video_id
user_id
video_type
content_text
media_url
thumbnail_url
like_counts
share_count
comment_count
other meta data
Video DB
MetaData Storage for Video
Write Ops Fast
Pre Signed URL
Video upload in S3 bucket
AI Based Video checking
blocked_Videos
filtered_video
Video against policy
Notification SVC
Moderator SVC
KAFKA Producer
Video Consumer SVC
Filtering Videos
Video Upload Service
Check the Video's from Video DB in prior & pass them to KAFKA queue.
Video Materializer
Users_Latest_Videos
Helps in feed generation
Fanout Svc (PUSH)
< userID, List <videoID> >
KAFKA
For a specific user storing all video's to show on feed.
< video, List<subscribersUserID> >
Fanout Consumer
3. write
BackFill


(Top People (whose video we see) other's who see our videos)
recommendation for top models
2. Get the list of all subscribers to backfill
Read
After live ends, video is stored in normal video DB
SUBSCRIBER SERVICE
Subscriber Service
Subscribers
Subscriber_id (PK)
Subscribers_id
Notifications(s8, mute)
Timestamp
User Engagement Service
Redis Cache:
For like/comment counts etc
Write heavy operations
Comment
comment_id
post_id
user_id
content
other data
Like
like_id
post_id / comment_id
user_id
reaction_type
other data
LIVE STREAM SERVICE
Creator App
RTMP INGESTION SERVER
RTMP: Real Time Messaging Protocol
Acts as the initial entry point for data, receiving, processing, and routing incoming information
Real Time Transcoder
Segmenter
Live Manifest File generation
HLS or DASH
BLOB Storage
Live Media Storage
RTMP: For sending video to server
HLS/DASH: for video delivery
Load Balancer For WS
Web Socket Gateway & Load Balancers
Routing
Rate Limiting
Viewer Side
CDN
User 1 connected to Live chat service 1
For storing likes and comments
Live Chat SVC 1
User 4 connected to live chat service 2
Live Chat SVC 2
Live Chat SVC 3
user 1: web-socket1 , TTL
user 2: web-socket2 , TTL
user 3: web-socket3 , TTL
these are open WS connections
web-socket registry stored in REDIS

Outcome

- Designed a highly scalable, event-driven video streaming architecture.
- Identified critical functional and non-functional requirements for media delivery.
- Designed REST APIs and decoupled asynchronous processing pipelines.
- Understood the application of CDNs, adaptive bitrate streaming, and stream processing for massive-scale systems.